

Exact Dyadic Extrapolation for Finite Rvachev Sinc-Product Splines

Eventual geometric modes, finite deconvolution, and a single quarter-base Gaussian-binomial formula

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Abstract

Let

$$\widehat{\text{up}}(t) = \prod_{k=0}^{\infty} \text{sinc}\left(\frac{\pi t}{2^k}\right), \quad \widehat{\text{up}}_n(t) = \prod_{k=0}^n \text{sinc}\left(\frac{\pi t}{2^k}\right),$$

under the Fourier convention $\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i t x} dx$ and $\text{sinc } z = \sin z / z$. The inverse transform up_n is a compactly supported piecewise-polynomial box spline of degree n . Fix a reduced dyadic point $x = a/2^r \in [-1, 1]$ and put $m = \lfloor r/2 \rfloor$ and $\rho = 1/4$. We prove the exact, not merely asymptotic, identity

$$\text{up}_n(x) = \text{up}(x) + \sum_{\ell=1}^m \frac{\gamma_{2\ell}}{(2\ell)!} \text{up}^{(2\ell)}(x) \rho^{\ell(n+1)}, \quad n \geq r,$$

where $1/M(z) = \sum_{j \geq 0} \gamma_j z^j / j!$ and $M(z) = \int_{\mathbb{R}} e^{zx} \text{up}(x) dx$ is the moment generating function of the up-density. Thus the error consists of at most m geometric modes, with successive ratios $4^{-1}, 4^{-2}, \dots, 4^{-m}$. The onset $n = r$ is explained by the knot geometry: x becomes the center of a polynomial cell precisely at that level, whereas at level $r - 1$ it is a knot.

Writing $q_n = \text{up}_n(x)$, the true value is recovered from any $m + 1$ consecutive values beginning at $N \geq r$ by the single exact formula

$$\text{up}(x) = \sum_{j=0}^m \frac{(-1)^{m-j} \rho^{\binom{m-j+1}{2}}}{(\rho; \rho)_j (\rho; \rho)_{m-j}} q_{N+j} = \frac{1}{(\rho; \rho)_m} \sum_{j=0}^m (-1)^{m-j} \rho^{\binom{m-j+1}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_{\rho} q_{N+j}.$$

All quantities are rational. The article gives complete proofs, a formula for all correction coefficients, an annihilating recurrence, a stable recursive Richardson tableau, exact examples, and a standard-library Python verifier using rational arithmetic only.

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1 Introduction and the main result

The Rvachev up-function is the compactly supported C^∞ function on $[-1, 1]$ characterized by

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1), \quad \text{up}(0) = 1. \quad (1)$$

Its Fourier transform is the geometrically scaled sinc product displayed in the abstract. These equivalent viewpoints, together with its probabilistic interpretation and rationality at dyadic points, go back to the classical work summarized by Arias de Reyna [1, 2]; see also Fabius [5] and Haugland [7].

Truncating the Fourier product after $k = n$ replaces the smooth up-function by a nonuniform box spline up_n . At a dyadic point each $q_n = \text{up}_n(x)$ is rational, and $q_n \rightarrow \text{up}(x)$. The question addressed here is stronger than convergence: what is the exact form of the tail of the sequence, and can its limit be extracted by one finite rational formula?

The answer is controlled by three facts.

1. The omitted Fourier tail is not an arbitrary perturbation. It is an exactly rescaled copy of the full up-law, of spatial radius $\delta_n = 2^{-(n+1)}$.
2. If $x = a/2^r$ is reduced, then for every $n \geq r$ the interval $(x - \delta_n, x + \delta_n)$ lies in one polynomial cell of up_n .
3. The Taylor jet of up at x terminates at order r : $\text{up}^{(j)}(x) = 0$ for $j > r$. Since the omitted tail is symmetric, only even orders survive.

The convolution equation can therefore be inverted on a finite-dimensional jet space. The inverse is the reciprocal moment-generating series of the up-law, and its powers of $\delta_n^2 = 4^{-(n+1)}$ are precisely the desired geometric modes.

Main theorem in operational form. Let $x = a/2^r \in [-1, 1]$ be in lowest terms, let $m = \lfloor r/2 \rfloor$, put $\rho = 1/4$, and set $q_n = \text{up}_n(x)$. Then for every $n \geq r$ there are rational constants $A_1(x), \dots, A_m(x)$ such that

$$q_n = \text{up}(x) + \sum_{\ell=1}^m A_\ell(x) \rho^{\ell(n+1)}, \quad A_\ell(x) = \frac{\gamma^{2\ell}}{(2\ell)!} \text{up}^{(2\ell)}(x).$$

For any $N \geq r$, the $m + 1$ samples $q_N, \dots, q_{N+m}$ determine the limit and every A_ℓ . In particular, the boxed q-binomial formula in the abstract returns $\text{up}(x)$ exactly.

This is geometric Richardson extrapolation, but in an unusually rigid form: the error expansion terminates. No limiting process and no remainder bound are needed once $n \geq r$. The Gaussian-binomial row is the closed form of Lagrange interpolation at the geometric nodes $1, \rho, \dots, \rho^m$. The denominator-free finite q-binomial and geometric Lagrange identities used in this step agree with the algebraic infrastructure already developed in the Provel repository [8]; the analytic bridge that identifies the particular spline sequence with a finite quarter-base mode expansion is proved below from first principles.

2 Fourier normalization and the finite box splines

2.1 Every sinc factor is a centered uniform density

For $a > 0$, define the centered box density

$$b_a(x) = \frac{1}{a} \mathbf{1}_{[-a/2, a/2]}(x). \quad (2)$$

A direct integration gives

$$\widehat{b}_a(t) = \frac{\sin(\pi at)}{\pi at} = \text{sinc}(\pi at). \quad (3)$$

Consequently, if

$$b_k(x) := 2^k \mathbf{1}_{[-2^{-k-1}, 2^{-k-1}]}(x), \quad (4)$$

then $\widehat{b}_k(t) = \text{sinc}(\pi t/2^k)$.

Proposition 2.1 (Finite convolution model). *For every $n \geq 0$,*

$$\text{up}_n = b_0 * b_1 * \cdots * b_n. \quad (5)$$

It is a nonnegative probability density, is even, has support

$$\text{supp up}_n = [-1 + \delta_n, 1 - \delta_n], \quad \delta_n := 2^{-(n+1)}, \quad (6)$$

and for $n \geq 1$ is a C^{n-1} piecewise polynomial of degree n .

Proof. Equation (5) follows from (3), multiplication of Fourier transforms, and uniqueness of inversion. Convolution preserves nonnegativity, total mass one, and evenness. The support radius is the sum of the box half-widths,

$$\sum_{k=0}^n 2^{-k-1} = 1 - 2^{-(n+1)} = 1 - \delta_n.$$

Convolving $n + 1$ interval indicators gives a univariate box spline of degree n and smoothness C^{n-1} because all widths are positive and the knots are simple after accounting for coincident subset sums; the explicit formula in [theorem 2.2](#) also proves the polynomial statement directly. $\square$

Probabilistically, up_n is the density of

$$Z_n = X_0 + \cdots + X_n, \quad X_k \sim \text{Unif}[-2^{-k-1}, 2^{-k-1}], \quad (7)$$

with the X_k independent. The full up-density is the law of the almost surely convergent sum $Z = \sum_{k \geq 0} X_k$.

2.2 An exact truncated-power formula

For $y \in \mathbb{R}$ and integer $d \geq 1$, write $y_+^d = (\max\{y, 0\})^d$. The standard inclusion–exclusion formula for a sum of independent nonidentically distributed uniforms becomes especially transparent for dyadic widths.

Theorem 2.2 (Exact finite spline formula). *For $n \geq 1$,*

$$\text{up}_n(x) = \frac{2^{\binom{n+1}{2}}}{n!} \sum_{j=0}^{2^{n+1}-1} (-1)^{s_2(j)} \left(x + 1 - 2^{-n-1} - \frac{j}{2^n} \right)_+^n, \quad (8)$$

where $s_2(j)$ is the sum of the binary digits of j . In particular, $\text{up}_n(x) \in \mathbb{Q}$ whenever x is dyadic.

Proof. Let $a_k = 2^{-k}$ and $A = \sum_{k=0}^n a_k = 2 - 2^{-n}$. Translate each X_k in (7) to a uniform variable on $[0, a_k]$. Repeated integration of the indicator of the simplex $\{y_k \geq 0 : \sum y_k \leq t\}$, followed by inclusion-exclusion for the upper bounds $y_k \leq a_k$, gives the density

$$\frac{1}{n! \prod_{k=0}^n a_k} \sum_{S \subseteq \{0, \dots, n\}} (-1)^{|S|} \left(x + \frac{A}{2} - \sum_{k \in S} a_k \right)_+^n. \quad (9)$$

Now

$$\prod_{k=0}^n a_k = 2^{-(n+1)}, \quad \frac{A}{2} = 1 - 2^{-n-1}.$$

The map $S \mapsto j = 2^n \sum_{k \in S} 2^{-k}$ is a bijection from subsets of $\{0, \dots, n\}$ to the integers $0 \leq j < 2^{n+1}$, and $|S| \equiv s_2(j) \pmod{2}$. Substitution into (9) proves (8). Every summand is rational at a dyadic x . $\square$

For $n = 0$, up_0 is the unit-height box on $[-1/2, 1/2]$, with the usual half-height Fourier inversion convention at the two jump points. All results below either begin at $n \geq 1$ or are trivial at level zero.

2.3 The knot lattice and the exact onset level

The possible breakpoints in (8) are

$$-1 + \delta_n + \frac{j}{2^n} = (-2^{n+1} + 1 + 2j)\delta_n, \quad 0 \leq j < 2^{n+1}. \quad (10)$$

Thus all knots are odd multiples of δ_n , consecutive knots are $2\delta_n$ apart, and the centers of the polynomial cells are the even multiples of δ_n .

Definition 2.3 (Reduced dyadic level). *For a dyadic rational $x \in [-1, 1]$, its reduced level $r = r(x)$ is defined by $x = a/2^r$ in lowest terms. Thus a is odd when $r > 0$, while the integers $-1, 0, 1$ have level 0.*

Proposition 2.4 (Cell-center stabilization). *Let $x = a/2^r$ be reduced.*

1. *If $|x| < 1$, then for every $n \geq r$ the point x is the center of one polynomial cell of up_n , and that cell is exactly $(x - \delta_n, x + \delta_n)$.*
2. *If $x = \pm 1$, then up_n is identically zero on the open interval $(x - \delta_n, x + \delta_n)$, apart from the support endpoint at one boundary. Thus the same interval is governed by the single zero polynomial for purposes of convolution and jet evaluation.*
3. *If $r > 0$ (hence $|x| < 1$), then at level $n = r - 1$ the point x is a knot.*

Proof. Since $\delta_n = 2^{-n-1}$,

$$\frac{x}{\delta_n} = a2^{n+1-r}. \quad (11)$$

For $n \geq r$ this is even. If $|x| < 1$, the point is therefore an even multiple of δ_n lying between the extreme knots, hence a cell center. Its adjacent knots are the neighboring odd multiples, at distance δ_n . For $x = \pm 1$, the support formula (6) gives the stated exterior zero region. Finally, at $n = r - 1$ the quotient in (11) equals the odd integer a , so an interior x is a knot. $\square$

This proposition explains both the phrase “except for a few initial values” and the precise number of exceptions: $n < r$ is the only range in which the small tail window can cross one or more internal knots. The guarantee $n \geq r$ is sharp at the level of local geometry. Special points can exhibit accidental earlier agreement, but no such accident is needed or assumed.

3 The omitted tail is a rescaled copy of the full law

The infinite product splits as

$$\widehat{\text{up}}(t) = \widehat{\text{up}}_n(t) \prod_{k=n+1}^{\infty} \text{sinc}\left(\frac{\pi t}{2^k}\right) \quad (12)$$

$$= \widehat{\text{up}}_n(t) \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi(2^{-n-1}t)}{2^j}\right) \quad (13)$$

$$= \widehat{\text{up}}_n(t) \widehat{\text{up}}(\delta_n t). \quad (14)$$

For $\delta > 0$, let

$$T_\delta(y) = \delta^{-1} \text{up}(y/\delta). \quad (15)$$

Then $\widehat{T_\delta}(t) = \widehat{\text{up}}(\delta t)$.

Lemma 3.1 (Exact self-similar tail). *For every $n \geq 0$,*

$$\boxed{\text{up} = \text{up}_n * T_{\delta_n}, \quad \delta_n = 2^{-n-1}.} \quad (16)$$

Equivalently, if Z' is an independent copy of the full random variable Z , then

$$Z \stackrel{d}{=} Z_n + \delta_n Z'. \quad (17)$$

Proof. Equation (16) is Fourier inversion applied to (14). For the random-variable form, reindex the omitted sum:

$$\sum_{k=n+1}^{\infty} X_k \stackrel{d}{=} 2^{-n-1} \sum_{j=0}^{\infty} X'_j = \delta_n Z',$$

where X'_j has the same centered uniform law as X_j . □

This exact renormalization is the source of the geometric parameter. The spatial scale of the tail is δ_n , and symmetry will force errors to use only even powers, hence the basic ratio

$$\rho := \delta_{n+1}^2 / \delta_n^2 = \frac{1}{4}. \quad (18)$$

4 Finite dyadic jets of the up-function

The second ingredient is the exact termination of the Taylor jet at a dyadic point. We prove it in a form that also makes all derivative values explicitly computable from up-values.

Define the Thue–Morse signs

$$\varepsilon_j = (-1)^{s_2(j)}. \quad (19)$$

Theorem 4.1 (Thue–Morse derivative stencil). *For every integer $d \geq 0$ and every $x \in \mathbb{R}$,*

$$\boxed{\text{up}^{(d)}(x) = 2^{\binom{d+1}{2}} \sum_{j=0}^{2^d-1} \varepsilon_j \text{up}(2^d x + 2^d - 1 - 2j).} \quad (20)$$

Proof. For $d = 0$ the assertion is tautological. The case $d = 1$ is exactly (1). Suppose the formula holds for d . Differentiate and use the chain rule:

$$\begin{aligned} \text{up}^{(d+1)}(x) &= 2^{\binom{d+1}{2}} 2^d \sum_{j < 2^d} \varepsilon_j \text{up}'(2^d x + 2^d - 1 - 2j) \\ &= 2^{\binom{d+1}{2} + d + 1} \sum_{j < 2^d} \varepsilon_j [\text{up}(2^{d+1} x + 2^{d+1} - 1 - 4j) \\ &\quad - \text{up}(2^{d+1} x + 2^{d+1} - 3 - 4j)]. \end{aligned}$$

The first term is indexed by $2j$ and the second by $2j + 1$. Since $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$, the two sums combine into (20) with $d + 1$. Finally, $\binom{d+1}{2} + d + 1 = \binom{d+2}{2}$. $\square$

Corollary 4.2 (Dyadic jet truncation). *If $x = a/2^r$ is reduced, then*

$$\text{up}^{(d)}(x) = 0 \quad (d > r). \quad (21)$$

For an interior point $-1 < x < 1$, the top derivative is nonzero:

$$\text{up}^{(r)}(a/2^r) = 2^{\binom{r+1}{2}} (-1)^{s_2((a+2^r-1)/2)}. \quad (22)$$

Proof. If $d > r$, then $2^d x = a2^{d-r}$ is even, while $2^d - 1 - 2j$ is odd. Every argument in (20) is therefore an odd integer. The only odd integers in $[-1, 1]$ are ± 1 , where up vanishes; all other terms are outside the support. Hence the sum is zero.

For $d = r$, all arguments are even integers. Exactly one can lie in $[-1, 1]$, namely zero, obtained at $j = (a + 2^r - 1)/2$. This index lies in $\{0, \dots, 2^r - 1\}$ because $-2^r < a < 2^r$. Since $\text{up}(0) = 1$, formula (22) follows. $\square$

The odd-level case needs one further observation when we later prove that the largest even mode really occurs.

Proposition 4.3 (The highest even derivative is nonzero). *Let $x = a/2^r \in (-1, 1)$ be reduced and set $m = \lfloor r/2 \rfloor$. If $r \geq 2$, then*

$$\text{up}^{(2m)}(x) \neq 0. \quad (23)$$

More precisely,

$$\left| \text{up}^{(2m)}(x) \right| = \begin{cases} 2^{\binom{r+1}{2}}, & r = 2m, \\ 2^{\binom{r}{2}-1}, & r = 2m + 1. \end{cases} \quad (24)$$

Proof. For even r , this is (22). Let $r = 2m + 1$ be odd and at least 3, and put $d = r - 1 = 2m$. The arguments in (20) are half-integers. The only half-integers in $[-1, 1]$ are $\pm 1/2$. Exactly one of the two equations

$$\frac{a}{2} + 2^{r-1} - 1 - 2j = \frac{1}{2}, \quad \frac{a}{2} + 2^{r-1} - 1 - 2j = -\frac{1}{2}$$

has an integral solution j : the two candidate numerators differ by 2, and $a + 2^r$ is odd. The valid solution lies between 0 and $2^{r-1} - 1$. Since $\text{up}(1/2) = \text{up}(-1/2) = 1/2$, the stencil contains one nonzero summand of magnitude $1/2$. Multiplication by $2^{\binom{d+1}{2}} = 2^{\binom{r}{2}}$ yields the second case of (24). $\square$

5 The reciprocal moment series and its q-product structure

5.1 Moment and cumulant generating functions

Let Z have density up . Its moment generating function is

$$M(z) := \mathbb{E}e^{zZ} = \int_{-1}^1 e^{zx} \text{up}(x) dx = \prod_{k=0}^{\infty} \frac{\sinh(z/2^{k+1})}{z/2^{k+1}}. \quad (25)$$

It is an even entire function with $M(0) = 1$. Write

$$M(z) = \sum_{j=0}^{\infty} \mu_j \frac{z^j}{j!}, \quad \frac{1}{M(z)} = \sum_{j=0}^{\infty} \gamma_j \frac{z^j}{j!}. \quad (26)$$

Thus $\mu_j = \mathbb{E}Z^j$, and all odd μ_j and γ_j vanish. The first moments are

$$\mu_0 = 1, \quad \mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}. \quad (27)$$

Let κ_j be the cumulants, so that

$$\log M(z) = \sum_{j=1}^{\infty} \kappa_j \frac{z^j}{j!}. \quad (28)$$

Proposition 5.1 (Rational cumulants and reciprocal coefficients). *All odd cumulants vanish, and for $s \geq 1$,*

$$\boxed{\kappa_{2s} = \frac{B_{2s}}{2s(1 - 4^{-s})}}, \quad (29)$$

where B_{2s} is a Bernoulli number. The reciprocal coefficients satisfy

$$\gamma_0 = 1, \quad \gamma_n = - \sum_{j=1}^n \binom{n-1}{j-1} \kappa_j \gamma_{n-j}. \quad (30)$$

In particular, every γ_j is rational.

Proof. For $V \sim \text{Unif}[-1, 1]$, $\mathbb{E}e^{zV} = \sinh z/z$ and

$$\log \frac{\sinh z}{z} = \sum_{s \geq 1} \frac{2^{2s-1} B_{2s}}{s(2s)!} z^{2s}. \quad (31)$$

Since $Z = \sum_{k \geq 0} 2^{-k-1} V_k$ with independent copies V_k , cumulants add under convolution and scale homogeneously. Multiplying the cumulant of V by $\sum_{k \geq 0} 2^{-2s(k+1)} = (2^{2s} - 1)^{-1}$ gives (29). Finally, differentiate $M(z)M(z)^{-1} = 1$ in exponential-generating-function form, or use the standard complete Bell-polynomial recurrence, to obtain (30). $\square$

The values relevant to the first six correction modes are shown in [table 1](#).

Table 1: Reciprocal moment coefficients.

j	γ_j	$\gamma_j/j!$
0	1	1
2	$-1/9$	$-1/18$
4	$31/675$	$31/16200$
6	$-2347/59535$	$-2347/42865200$
8	$1904369/32531625$	$1904369/1311675120000$
10	$-3309760579/24405225075$	$-3309760579/88561680752160000$
12	1884491755011469	1884491755011469
	4133856862760625	$1980124051433319792000000$

5.2 Euler products and q-Pochhammer coefficients

Put $\rho = 1/4$ and use

$$(a; \rho)_N := \prod_{j=0}^{N-1} (1 - a\rho^j), \quad (a; \rho)_\infty := \prod_{j=0}^{\infty} (1 - a\rho^j). \quad (32)$$

Euler's product for sinh gives

$$M(z) = \prod_{k=0}^{\infty} \prod_{h=1}^{\infty} \left(1 + \frac{z^2}{4\pi^2 h^2 4^k} \right) \quad (33)$$

$$= \prod_{h=1}^{\infty} \left(-\frac{z^2}{4\pi^2 h^2}; \rho \right)_{\infty}. \quad (34)$$

Hence

$$\frac{1}{M(z)} = \prod_{h=1}^{\infty} \frac{1}{(-z^2/(4\pi^2 h^2); \rho)_{\infty}}. \quad (35)$$

The special case of the q-binomial theorem

$$\frac{1}{(a; \rho)_{\infty}} = \sum_{j=0}^{\infty} \frac{a^j}{(\rho; \rho)_j} \quad (36)$$

then supplies a direct q-Pochhammer expression for every reciprocal moment coefficient.

Proposition 5.2 (q-Pochhammer coefficient formula). *For $s \geq 0$,*

$$\frac{\gamma_{2s}}{(2s)!} = \frac{(-1)^s}{(4\pi^2)^s} \sum_{\substack{\nu_1, \nu_2, \dots \geq 0 \\ \nu_1 + \nu_2 + \dots = s}} \prod_{h=1}^{\infty} \frac{h^{-2\nu_h}}{(\rho; \rho)_{\nu_h}}. \quad (37)$$

The summation is over finitely supported sequences. In particular, $(-1)^s \gamma_{2s} > 0$, so no even reciprocal coefficient vanishes.

Proof. Insert (36) into each factor of (35) and collect the coefficient of z^{2s} . Every contributing monomial has total exponent $\sum_h \nu_h = s$ and hence common sign $(-1)^s$. After that sign is removed, all summands are strictly positive. Absolute convergence near the origin justifies coefficient collection analytically; alternatively the identity is valid coefficientwise as a convergent formal product because the sums contain zeta-type factors $\sum h^{-2j}$. $\square$

Formula (37) is useful conceptually: the same quarter-base q -calculus controls both the deconvolution coefficients and the finite extrapolation row. Rationality is more transparent from (29)–(30); alternating nonvanishing is more transparent from (37).

6 Finite local deconvolution

We now invert the convolution (16) at a cell center. The following finite-dimensional lemma is the analytic core of the article.

Lemma 6.1 (Finite jet inversion). *Let P be a polynomial of degree at most n , let $\delta > 0$, and suppose a smooth function f has, at a point x , the jet relations*

$$f^{(j)}(x) = \int_{-1}^1 P^{(j)}(x - \delta z) \text{up}(z) \, dz, \quad 0 \leq j \leq n. \quad (38)$$

Then for $0 \leq j \leq n$,

$$P^{(j)}(x) = \sum_{k=0}^{n-j} \frac{\gamma_k}{k!} \delta^k f^{(j+k)}(x). \quad (39)$$

Proof. Taylor expansion is finite, so symmetry of up gives

$$f^{(j)}(x) = \sum_{k=0}^{n-j} \frac{\mu_k}{k!} \delta^k P^{(j+k)}(x). \quad (40)$$

(The sign from $x - \delta z$ is immaterial because odd moments vanish.) Substitute (40) into the right side of (39). The coefficient multiplying $\delta^s P^{(j+s)}(x)$ is

$$\sum_{k=0}^s \frac{\gamma_k \mu_{s-k}}{k!(s-k)!} = [z^s] M(z)^{-1} M(z).$$

It equals 1 when $s = 0$ and 0 otherwise. All sums are finite, so no convergence issue arises. $\square$

Theorem 6.2 (Exact local deconvolution of a sinc prefix). *Let $x = a/2^r \in [-1, 1]$ be reduced and $n \geq r$. Then*

$$\text{up}_n(x) = \sum_{k=0}^r \frac{\gamma_k}{k!} \delta_n^k \text{up}^{(k)}(x) = \sum_{\ell=0}^{\lfloor r/2 \rfloor} \frac{\gamma_{2\ell}}{(2\ell)!} \delta_n^{2\ell} \text{up}^{(2\ell)}(x). \quad (41)$$

The same identity with x replaced by a nearby point need not hold: its validity is tied to the single polynomial cell centered at x .

Proof. By theorem 2.4, the restriction of up_n to $(x - \delta_n, x + \delta_n)$ is one polynomial P of degree at most n . Using (16) and the substitution $y = \delta_n z$ gives

$$\text{up}(x) = \int_{-1}^1 P(x - \delta_n z) \text{up}(z) \, dz. \quad (42)$$

Distributional differentiation of the convolution, followed by restriction to the open cell, gives (38) for $0 \leq j \leq n$. At order n , the piecewise-constant derivative may jump at the two cell endpoints, but

those endpoints correspond to $z = \pm 1$, a measure-zero set where the tail density also vanishes. Thus [theorem 6.1](#) applies with $f = \text{up}$, $\delta = \delta_n$, and $j = 0$:

$$\text{up}_n(x) = P(x) = \sum_{k=0}^n \frac{\gamma_k}{k!} \delta_n^k \text{up}^{(k)}(x).$$

Terms with $k > r$ vanish by [theorem 4.2](#); odd terms vanish because $\gamma_k = 0$ for odd k . $\square$

The theorem can also be read as an exact differential-operator identity on the finite jet:

$$\text{up}_n(x) = [M(\delta_n D)^{-1} \text{up}](x), \quad (43)$$

where the formal inverse automatically truncates because the dyadic jet is finite.

7 The eventual finite geometric expansion

Set

$$q_n := \text{up}_n(x), \quad \rho := \frac{1}{4}, \quad m := \lfloor r/2 \rfloor. \quad (44)$$

Since $\delta_n^{2\ell} = \rho^{\ell(n+1)}$, the desired structure now follows immediately.

Theorem 7.1 (Eventual geometric-mode theorem). *For a reduced dyadic $x = a/2^r \in [-1, 1]$, define*

$$A_\ell(x) := \frac{\gamma_{2\ell}}{(2\ell)!} \text{up}^{(2\ell)}(x), \quad 1 \leq \ell \leq m = \lfloor r/2 \rfloor. \quad (45)$$

Then, for every $n \geq r$,

$$q_n = \text{up}(x) + \sum_{\ell=1}^m A_\ell(x) \rho^{\ell(n+1)}, \quad \rho = \frac{1}{4}. \quad (46)$$

Each summand is a geometric progression in n with ratio $4^{-\ell}$. All coefficients $A_\ell(x)$ are rational.

Proof. Equation (46) is (41) with the constant term separated. It remains to justify rationality without assuming the classical dyadic-value theorem. The finite spline formula (8) shows that every q_n is rational. In [section 8](#) we construct rational weights that annihilate all nonconstant modes in (46); hence those weights express $\text{up}(x)$ as a rational combination of finitely many q_n , proving $\text{up}(x) \in \mathbb{Q}$. Formula (20) then expresses each $\text{up}^{(2\ell)}(x)$ as an integer multiple of finitely many up-values at dyadic points, all rational by the same argument. Finally, $\gamma_{2\ell} \in \mathbb{Q}$ by [theorem 5.1](#). $\square$

There is no circularity in the rationality argument: the geometric expansion is first established over $\mathbb{R}$; the extraction row then proves that its constant term is rational.

Corollary 7.2 (Sharp maximal mode). *If $x \in (-1, 1)$ has reduced level $r \geq 2$, then the coefficient of the fastest listed mode, $A_m(x)$, is nonzero. Hence the right side of (46), regarded as a polynomial in $z = \rho^{n+1}$, has exact degree m .*

Proof. By [theorem 5.2](#), $\gamma_{2m} \neq 0$. By [theorem 4.3](#), $\text{up}^{(2m)}(x) \neq 0$. $\square$

Intermediate A_ℓ may be treated uniformly whether or not any special cancellation occurs. The theorem does not assert that the sequence q_n itself is monotone; it asserts that each constituent error mode decays monotonically in magnitude.

Corollary 7.3 (Derivative version). *Let $0 \leq s \leq r$ and $n \geq \max\{r, s\}$. Then*

$$\text{up}_n^{(s)}(x) = \text{up}^{(s)}(x) + \sum_{\ell=1}^{\lfloor (r-s)/2 \rfloor} \frac{\gamma_{2\ell}}{(2\ell)!} \text{up}^{(s+2\ell)}(x) \rho^{\ell(n+1)}. \quad (47)$$

Thus the same extraction method recovers dyadic derivative values, with $\lfloor (r-s)/2 \rfloor + 1$ samples.

Proof. Apply (39) with the chosen s instead of 0, then use (21) and parity of γ_j . $\square$

8 Exact recovery by one q-binomial formula

Fix $N \geq r$. Define the degree- m polynomial

$$R_N(z) := \text{up}(x) + \sum_{\ell=1}^m A_\ell(x) \rho^{\ell(N+1)} z^\ell. \quad (48)$$

Then the consecutive spline values are samples at geometric nodes:

$$q_{N+j} = R_N(\rho^j), \quad 0 \leq j \leq m, \quad (49)$$

and the desired value is $R_N(0)$.

Recall

$$(\rho; \rho)_j = \prod_{h=1}^j (1 - \rho^h), \quad \left[\begin{matrix} m \\ j \end{matrix} \right]_\rho = \frac{(\rho; \rho)_m}{(\rho; \rho)_j (\rho; \rho)_{m-j}}. \quad (50)$$

Theorem 8.1 (Single q-Pochhammer extraction formula). *Let $x = a/2^r$ be reduced, $m = \lfloor r/2 \rfloor$, and $N \geq r$. Then*

$$\text{up}(x) = \sum_{j=0}^m \frac{(-1)^{m-j} \rho^{\binom{m-j+1}{2}}}{(\rho; \rho)_j (\rho; \rho)_{m-j}} q_{N+j}, \quad \rho = \frac{1}{4}. \quad (51)$$

Equivalently,

$$\text{up}(x) = \frac{1}{(\rho; \rho)_m} \sum_{j=0}^m (-1)^{m-j} \rho^{\binom{m-j+1}{2}} \left[\begin{matrix} m \\ j \end{matrix} \right]_\rho q_{N+j}. \quad (52)$$

Both formulas are exact identities in $\mathbb{Q}$.

First proof: geometric Lagrange interpolation. Lagrange interpolation of R_N at the nodes $1, \rho, \dots, \rho^m$ gives

$$R_N(0) = \sum_{j=0}^m q_{N+j} \prod_{\substack{0 \leq k \leq m \\ k \neq j}} \frac{-\rho^k}{\rho^j - \rho^k}. \quad (53)$$

Split the product into $k < j$ and $k > j$. For $k < j$,

$$\frac{-\rho^k}{\rho^j - \rho^k} = \frac{1}{1 - \rho^{j-k}},$$

while for $k > j$,

$$\frac{-\rho^k}{\rho^j - \rho^k} = -\frac{\rho^{k-j}}{1 - \rho^{k-j}}.$$

The lower factors contribute $(\rho; \rho)_j^{-1}$. The upper factors contribute

$$\frac{(-1)^{m-j} \rho^{1+2+\dots+(m-j)}}{(\rho; \rho)_{m-j}} = \frac{(-1)^{m-j} \rho^{\binom{m-j+1}{2}}}{(\rho; \rho)_{m-j}}.$$

Substitution into (53) proves (51); multiplying and dividing by $(\rho; \rho)_m$ gives (52). $\square$

Second proof: an annihilating shift polynomial. Let E denote the forward shift, $(Eq)_n = q_{n+1}$. On the constant sequence 1, E has eigenvalue 1; on the ℓ th error mode $\rho^{\ell(n+1)}$, it has eigenvalue ρ^ℓ . Therefore

$$\prod_{\ell=1}^m (E - \rho^\ell) q_N = (\rho; \rho)_m \text{up}(x). \quad (54)$$

The finite q -binomial theorem gives

$$\prod_{\ell=1}^m (z - \rho^\ell) = \sum_{j=0}^m (-1)^{m-j} \rho^{\binom{m-j+1}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_\rho z^j. \quad (55)$$

Insert $z = E$ into (55), apply it to q_N , and divide (54) by $(\rho; \rho)_m$. $\square$

The first few quarter-base rows are particularly simple.

Table 2: Exact extraction rows at $\rho = 1/4$. The numerator vector multiplies $(q_N, \dots, q_{N+m})$.

m	numerator vector	denominator
0	(1)	1
1	(-1, 4)	3
2	(1, -20, 64)	45
3	(-1, 84, -1344, 4096)	2835
4	(1, -340, 22848, -348160, 1048576)	722925
5	(-1, 1364, -371008, 23744512, -357564416, 1073741824)	739552275

Thus, for example,

$$m = 1 : \quad \text{up}(x) = \frac{4q_{N+1} - q_N}{3}, \quad (56)$$

$$m = 2 : \quad \text{up}(x) = \frac{q_N - 20q_{N+1} + 64q_{N+2}}{45}. \quad (57)$$

8.1 A recursive Richardson tableau

The closed row can be evaluated without first forming its large integer numerators.

Algorithm 8.2 (Exact quarter-base tableau). Set $R_0(N) = q_N$. For $s = 1, \dots, m$, define

$$R_s(N) = \frac{R_{s-1}(N+1) - \rho^s R_{s-1}(N)}{1 - \rho^s}. \quad (58)$$

Then R_s has its first s geometric modes removed, and

$$R_m(N) = \text{up}(x) \quad (N \geq r). \quad (59)$$

Proof. If a term in $R_{s-1}(N)$ is proportional to $\rho^{\ell N}$, the numerator in (58) multiplies it by $\rho^\ell - \rho^s$. The mode $\ell = s$ disappears, while the constant mode is multiplied by $1 - \rho^s$ and restored by division. Induction proves the claim. $\square$

8.2 Stability of the extraction row

Although exact arithmetic is natural here, the formula is also well conditioned for approximate samples. Let $\lambda_{m,j}$ denote the weight in (51).

Proposition 8.3 (Exact ℓ^∞ noise amplification). *If $\tilde{q}_{N+j} = q_{N+j} + e_j$, then*

$$|\widetilde{\text{up}}(x) - \text{up}(x)| \leq \Lambda_m(\rho) \max_j |e_j|, \quad \Lambda_m(\rho) := \sum_{j=0}^m |\lambda_{m,j}|. \quad (60)$$

Moreover,

$$\Lambda_m(\rho) = \frac{(-\rho; \rho)_m}{(\rho; \rho)_m} = \prod_{h=1}^m \frac{1 + \rho^h}{1 - \rho^h}. \quad (61)$$

At $\rho = 1/4$, these constants increase to

$$\Lambda_\infty(1/4) = \frac{(-1/4; 1/4)_\infty}{(1/4; 1/4)_\infty} \approx 1.9692603536682694. \quad (62)$$

Proof. The first statement is the triangle inequality. Taking absolute values in (52), replacing $k = m - j$, and using the finite q-binomial theorem at $z = -\rho$ yields

$$\begin{aligned} \Lambda_m(\rho) &= \frac{1}{(\rho; \rho)_m} \sum_{k=0}^m \rho^{\binom{k+1}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_\rho \\ &= \frac{1}{(\rho; \rho)_m} \prod_{h=1}^m (1 + \rho^h). \end{aligned}$$

This is (61). The infinite product converges because $\sum_h \rho^h < \infty$. $\square$

The bounded condition number is a useful contrast with ordinary polynomial extrapolation at nearly colliding nodes. Here the nodes are geometrically separated and the target 0 lies just outside their accumulation direction.

9 Recovering every correction coefficient

The limit is only the constant coefficient of R_N . The $m + 1$ unknowns may be displayed directly as a geometric Vandermonde system. Put $C_\ell = A_\ell(x)\rho^{\ell(N+1)}$. Then

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \rho & \rho^2 & \cdots & \rho^m \\ 1 & \rho^2 & \rho^4 & \cdots & \rho^{2m} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \rho^m & \rho^{2m} & \cdots & \rho^{m^2} \end{pmatrix} \begin{pmatrix} \text{up}(x) \\ C_1 \\ C_2 \\ \vdots \\ C_m \end{pmatrix} = \begin{pmatrix} q_N \\ q_{N+1} \\ q_{N+2} \\ \vdots \\ q_{N+m} \end{pmatrix}. \quad (63)$$

Its determinant is $\prod_{0 \leq i < j \leq m} (\rho^j - \rho^i) \neq 0$. Lagrange interpolation is the closed-form inverse of this system and recovers the full polynomial, hence every mode.

Theorem 9.1 (Full coefficient reconstruction). *For $N \geq r$, define*

$$\mathcal{L}_{m,j}(z) := \prod_{\substack{0 \leq k \leq m \\ k \neq j}} \frac{z - \rho^k}{\rho^j - \rho^k}. \quad (64)$$

Then

$$R_N(z) = \sum_{j=0}^m q_{N+j} \mathcal{L}_{m,j}(z), \quad (65)$$

and, for $1 \leq \ell \leq m$,

$$A_\ell(x) = \rho^{-\ell(N+1)} \sum_{j=0}^m q_{N+j} [z^\ell] \mathcal{L}_{m,j}(z). \quad (66)$$

All reconstructed coefficients are rational.

Proof. Equation (65) is ordinary Lagrange interpolation at the distinct nodes ρ^j . Comparing (48) coefficientwise gives (66). Every node and every sample is rational. $\square$

In practice, one can solve the Vandermonde system, use (65), or use a Newton basis adapted to the geometric nodes. The attached Python program uses exact Lagrange interpolation because it is short and transparent.

9.1 An exact recurrence and a model certificate

The sequence has characteristic roots $1, \rho, \dots, \rho^m$. Therefore it obeys a constant-coefficient recurrence of order $m + 1$.

Corollary 9.2 (Annihilating recurrence). *For every $n \geq r$,*

$$\prod_{\ell=0}^m (E - \rho^\ell) q_n = 0. \quad (67)$$

Equivalently,

$$\sum_{j=0}^{m+1} (-1)^{m+1-j} \rho^{\binom{m+1-j}{2}} \begin{bmatrix} m+1 \\ j \end{bmatrix}_\rho q_{n+j} = 0. \quad (68)$$

Proof. Each factor in (67) kills one of the $m + 1$ modes. The expanded form is the finite q-binomial theorem applied to $\prod_{\ell=0}^m (z - \rho^\ell)$. $\square$

This recurrence is useful in three ways. It certifies exact calculations; it can extend the sequence after $m + 1$ starting values are known; and it can empirically locate the onset if the denominator level was not supplied. For a known reduced dyadic point, however, no search is needed: $N = r$ is a proved safe choice.

Remark 9.3 (Why initial values can fail). At $n = r - 1$, x is a knot rather than a cell center. The convolution window $x - \delta_n z$ then samples two adjacent polynomial pieces of up_n . There is no single polynomial P to which [theorem 6.1](#) can be applied, and the recurrence (68) need not hold. At still earlier levels the window can meet more pieces. This is a geometric obstruction, not a weakness of the extrapolation algebra.

10 An exact computational method

For completeness, the entire extraction can be performed without evaluating up or any of its derivatives.

Algorithm 10.1 (Dyadic value extraction from finite sinc prefixes). *Input a dyadic $x \in [-1, 1]$.*

1. Reduce $x = a/2^r$ and set $m = \lfloor r/2 \rfloor$, $\rho = 1/4$, and $N = r$.
2. Compute the $m + 1$ rational spline values $q_N, q_{N+1}, \dots, q_{N+m}$, for example from (8).
3. Return (51) or, equivalently, apply the recursive tableau (58).
4. To recover the error constants as well, interpolate R_N by (65) and use (66).
5. Optionally verify additional levels with (46) or the recurrence (68).

The direct truncated-power evaluator contains 2^{n+1} terms and is intended as a reference implementation. It has exponential arithmetic cost in n , although an early cutoff reduces work near the left endpoint. The extraction stage itself uses only $m + 1 = \lfloor r/2 \rfloor + 1$ samples and $\mathcal{O}(m^2)$ elementary rational operations in a tableau, or $\mathcal{O}(m)$ operations after weights have been precomputed. Faster bit-recursive exact evaluators for the individual q_n can be substituted without changing any theorem.

10.1 Pseudocode

Listing 1: Core exact extraction in pseudocode.

```

rho = Fraction(1, 4)
r = log2(reduced_denominator(x))
m = r // 2
N = r
samples = [up_prefix_value(x, N + j) for j in range(m + 1)]

limit = 0
for j, qNj in enumerate(samples):
    weight = ((-1)**(m-j) * rho**binom(m-j+1, 2)
              / (qpoch(rho, j) * qpoch(rho, m-j)))
    limit += weight * qNj

```

The complete file `dyadic_up_extraction.py` adds exact Bernoulli and cumulant recurrences, derivative stencils, full mode interpolation, recurrence checks, command-line handling, and a suite of examples.

11 Worked exact examples

11.1 Level two: $x = 1/4$

Here $r = 2$, $m = 1$, and

$$q_n = \text{up}(1/4) + A_1(1/4)4^{-(n+1)}. \quad (69)$$

The derivative stencil gives $\text{up}''(1/4) = -8$, while $\gamma_2/2! = -1/18$, so $A_1 = 4/9$. Direct spline evaluation gives

$$q_2 = \frac{15}{16}, \quad q_3 = \frac{179}{192}. \quad (70)$$

The one-mode row (56) yields

$$\text{up}(1/4) = \frac{4q_3 - q_2}{3} = \frac{67}{72}. \quad (71)$$

Thus, for every $n \geq 2$,

$$\boxed{\text{up}_n(1/4) = \frac{67}{72} + \frac{4}{9} 4^{-(n+1)}}. \quad (72)$$

11.2 Level three: $x = 1/8$

Now $r = 3$ but still $m = 1$. We find $\text{up}''(1/8) = -4$, hence $A_1 = 2/9$, and

$$q_3 = \frac{383}{384}, \quad q_4 = \frac{1531}{1536}. \quad (73)$$

Therefore

$$\text{up}(1/8) = \frac{4q_4 - q_3}{3} = \frac{287}{288}, \quad \text{up}_n(1/8) = \frac{287}{288} + \frac{2}{9} 4^{-(n+1)} \quad (n \geq 3). \quad (74)$$

11.3 Level four: $x = 1/16$

Here $r = 4$ and $m = 2$, so two geometric modes are required. Exact derivative calculation gives

$$\text{up}''(1/16) = -\frac{5}{9}, \quad \text{up}^{(4)}(1/16) = -1024. \quad (75)$$

Using $\gamma_2/2! = -1/18$ and $\gamma_4/4! = 31/16200$,

$$A_1 = \frac{5}{162}, \quad A_2 = -\frac{3968}{2025}. \quad (76)$$

The first guaranteed samples are

$$q_4 = \frac{24575}{24576}, \quad q_5 = \frac{1965959}{1966080}, \quad q_6 = \frac{94365509}{94371840}. \quad (77)$$

The two-mode row (57) gives

$$\text{up}(1/16) = \frac{q_4 - 20q_5 + 64q_6}{45} = \frac{2073457}{2073600}. \quad (78)$$

The complete exact sequence tail is

$$\text{up}_n(1/16) = \frac{2073457}{2073600} + \frac{5}{162} 4^{-(n+1)} - \frac{3968}{2025} 16^{-(n+1)}, \quad n \geq 4. \quad (79)$$

At $n = 3$, the point $1/16$ is a knot and $q_3 = 1$; substituting $n = 3$ into (79) gives $46081/46080$, so the eventual formula genuinely fails before the proved onset.

11.4 A table of further values

Table 3: Exact values and geometric coefficients, with $q_n = \text{up}(x) + \sum_{\ell=1}^m A_\ell \rho^{\ell(n+1)}$.

x	r	m	$\text{up}(x)$	(A_1, A_2)
1/4	2	1	67/72	(4/9)
1/8	3	1	287/288	(2/9)
1/16	4	2	2073457/2073600	(5/162, -3968/2025)
3/16	4	2	2026943/2073600	(67/162, 3968/2025)
1/32	5	2	33177581/33177600	(1/648, -1984/2025)

x	r	m	$\text{up}(x)$	(A_1, A_2)
5/32	5	2	32842819/33177600	(215/648, 1984/2025)

Evenness gives the same values and mode coefficients at $-x$. At $x = 0$, $q_n = \text{up}(0) = 1$ for every n . At $x = \pm 1/2$, the reduced level is 1 and $m = 0$, so $q_n = \text{up}(x) = 1/2$ for every $n \geq 1$. At $x = \pm 1$, both the prefix and limiting values vanish.

12 Exact verification program

The attached program `dyadic_up_extraction.py` uses only Python's standard library and represents every number by `fractions.Fraction`. It performs the following independent calculations.

- It evaluates q_n directly from the 2^{n+1} -term spline formula (8).
- It computes the q-Pochhammer extraction weights and independently checks the Gaussian-binomial form.
- It interpolates $R_N(z)$ and extracts all A_ℓ .
- It computes Bernoulli numbers, cumulants, the reciprocal coefficients $\gamma_{2\ell}$, and dyadic derivatives through the Thue–Morse stencil.
- It verifies that the interpolated and differential-operator coefficients coincide exactly.
- It checks both the geometric expansion and the annihilating recurrence with zero rational residual.

A typical invocation is

```
python dyadic_up_extraction.py --x 1/16 --check-through 10
```

and the default invocation reproduces all rows in [tables 1 to 3](#). The companion file `verification_output.txt` records a complete default run. No numerical tolerance is used: every displayed PASS means equality of reduced fractions.

For example, the core output for $x = 1/16$ reads

```
x=1/16; r=4; m=2; up(x)=2073457/2073600
samples n=4..6: 24575/24576, 1965959/1966080,
                94365509/94371840
extraction row: [1, -20, 64]/45
modes A_ell: 5/162, -3968/2025
exact checks: PASS
```

The program is deliberately a transparent verifier rather than the fastest possible evaluator. It is useful for regression tests, for producing exact conjectural data at moderate levels, and as executable documentation of every formula in the paper.

13 Structural consequences

13.1 A finite-prefix proof of dyadic rationality

Classically, rationality of $\text{up}(a/2^r)$ follows from the differential functional equation and explicit recurrences [1, 2]. The present argument gives a different proof:

1. finite sinc prefixes are rational box splines at dyadic arguments;
2. their eventual tail belongs to a known finite-dimensional geometric mode space;
3. a rational q-Lagrange functional projects onto the constant mode.

Thus the limiting rational is obtained directly from finite convolution data. The proof is constructive and uses exactly $\lfloor r/2 \rfloor + 1$ consecutive prefix values once the cell-center level is reached.

13.2 Exactness versus asymptotics

Ordinary Richardson extrapolation begins from an asymptotic expansion such as $Q(h) = L + c_1 h^p + c_2 h^{p+1} + \dots$ and cancels a finite number of leading terms [9, 3]. Here the role of the mesh is $h = \delta_n$, but two extra facts make the result exact:

- the prefix is literally polynomial on the whole tail window;
- the target's dyadic jet terminates.

The reciprocal moment operator therefore has no uncomputed remainder. The method is better described as finite spectral projection than as approximate acceleration, although its coefficients are precisely geometric Richardson coefficients.

13.3 Minimal data within the mode model

For an interior point of level $r \geq 2$, the polynomial R_N has exact degree m by [theorem 7.2](#). Consequently, $m + 1$ generic values are the natural amount of data needed to recover its constant term when the remaining m coefficients are unknown. Fewer samples can succeed only by using additional point-specific information beyond the mode model, or through a special relation among coefficients. The universal q-binomial row is thus order-sharp for this finite-dimensional extrapolation problem.

13.4 A reusable local principle

The argument is not tied to sinc functions at every step. Suppose a smooth density f admits approximants p_n and scales δ_n such that

$$f = p_n * T_{\delta_n}, \quad T_{\delta}(y) = \delta^{-1} f(y/\delta),$$

and suppose p_n is one polynomial on the entire window $(x - \delta_n, x + \delta_n)$. If the jet of f at x vanishes above order r , then

$$p_n(x) = \sum_{k=0}^r b_k \delta_n^k f^{(k)}(x),$$

where $\sum b_k z^k$ is the reciprocal moment series of f . Whenever $\delta_n = c\alpha^n$, the sequence is a finite sum of geometric modes. The Rvachev setting is exceptionally clean because the tail is exactly self-similar, the knot grid aligns with dyadic denominators, the jet truncates, and symmetry removes every odd mode.

14 Relation to the ProveIt development

The linked ProveIt corpus contains several ingredients used here in more general or formally verified forms. The most relevant crosswalk is:

Table 4: Conceptual crosswalk to the repository.

Repository component	Role in the present article
<code>FiniteQBinomialCore.lean</code>	Denominator-free finite q-Pochhammer and Gaussian-binomial identities, including the finite q-binomial theorem used in (55).
<code>GeometricLagrangeQBinomial.lean</code>	Closed q-Pochhammer formulas for Lagrange weights on $1, q, \dots, q^m$; at $q = 1/4$ these are exactly the extraction weights in (51).
<code>QuarterCatalanRichardson.lean</code>	A formal quarter-base Richardson filter that preserves the constant term, cancels prescribed geometric degrees, and describes residual coefficients.
<code>Up_Polynomial_Synthesis.tex</code>	The up-law moment generating function, reciprocal Appell/deconvolution coefficients γ_j , Bernoulli cumulants, and finite sinc-prefix calculus.
<code>EarlyApproximants.lean</code>	Exact polynomial and measure approximants, coefficient normalization, and finite convolution infrastructure for related approximating families.

The present article isolates the particular approximants specified in the question – the direct Fourier prefixes $\prod_{k=0}^n \text{sinc}(\pi t/2^k)$ – and proves the missing bridge from their local cell geometry and the self-similar omitted tail to the finite geometric sequence model. The theorem and Python verification here should not be read as a claim that this exact bridge has already been formalized in Lean. A natural formalization plan is to combine:

1. a definition of the finite uniform convolution density up_n ;
2. its knot-cell polynomial theorem and the cell-center lemma;
3. a finite jet version of moment-series inversion over $\mathbb{Q}$;
4. the existing geometric Lagrange/q-binomial row;
5. the existing dyadic derivative stencil and rationality APIs.

This decomposition keeps the analytic measure/convolution layer separate from the purely algebraic extrapolation layer and should make theorem statements reusable for other self-similar kernels.

15 Conclusion

For a fixed reduced dyadic $x = a/2^r$, the finite sinc-prefix values $q_n = \text{up}_n(x)$ stop being an arbitrary convergent sequence at the explicit level $n = r$. From that point onward they lie exactly in the $(\lfloor r/2 \rfloor + 1)$ -dimensional span

$$1, \quad 4^{-(n+1)}, \quad 4^{-2(n+1)}, \quad \dots, \quad 4^{-\lfloor r/2 \rfloor(n+1)}.$$

The coefficients are not fitted approximations: they are the finite reciprocal-moment deconvolution coefficients $\gamma_{2\ell} \text{up}^{(2\ell)}(x)/(2\ell)!$. The mechanism is the exact self-similar omitted tail combined with the finite Taylor jet at a dyadic point.

The constant mode – the true rational value $\text{up}(x)$ – is selected by one quarter-base q-Pochhammer row, equivalently by a Gaussian-binomial Richardson filter. The same samples recover all geometric

coefficients, satisfy an exact q -binomial recurrence, and admit a bounded noise amplification factor $(-1/4; 1/4)_m / (1/4; 1/4)_m < 1.97$. This gives a complete exact answer to the extraction problem and a compact computational procedure suitable for direct use or formalization.

A Derivation of the general box-spline formula

For positive widths $a_0, \dots, a_n$, let Y_k be independent uniform variables on $[0, a_k]$ and let $S = \sum Y_k$. The volume

$$V(t) = \text{vol}\{y_k \geq 0 : y_k \leq a_k, \sum y_k \leq t\}$$

is obtained by translating every subset of violated upper bounds. The unbounded simplex $\{y_k \geq 0 : \sum y_k \leq t\}$ has volume $t_+^{n+1} / (n+1)!$, so inclusion-exclusion gives

$$V(t) = \frac{1}{(n+1)!} \sum_{S \subseteq \{0, \dots, n\}} (-1)^{|S|} \left(t - \sum_{k \in S} a_k \right)_+^{n+1}. \quad (80)$$

Divide by $\prod a_k$ and differentiate in t to get the density

$$f_S(t) = \frac{1}{n! \prod a_k} \sum_{S \subseteq \{0, \dots, n\}} (-1)^{|S|} \left(t - \sum_{k \in S} a_k \right)_+^n. \quad (81)$$

Centering each variable replaces t by $x + \frac{1}{2} \sum a_k$ and yields (9). This proof simultaneously establishes the piecewise-polynomial degree, support, and knot set.

B Finite q -binomial identities used in the extraction

The finite q -binomial theorem is

$$(z; q)_m = \prod_{h=0}^{m-1} (1 - zq^h) = \sum_{k=0}^m (-1)^k q^{\binom{k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q z^k. \quad (82)$$

Replacing z by zq gives

$$\prod_{h=1}^m (1 - zq^h) = \sum_{k=0}^m (-1)^k q^{\binom{k+1}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q z^k. \quad (83)$$

To obtain (55), write

$$\prod_{h=1}^m (z - q^h) = z^m \prod_{h=1}^m (1 - q^h/z)$$

and use (83) with z^{-1} , followed by the change of index $j = m - k$ and the symmetry $\begin{bmatrix} m \\ m-j \end{bmatrix}_q = \begin{bmatrix} m \\ j \end{bmatrix}_q$.

The cardinal weight at node q^j for evaluation at zero is

$$\lambda_{m,j} = \prod_{k \neq j} \frac{-q^k}{q^j - q^k} \quad (84)$$

$$= \frac{(-1)^{m-j} q^{1+\dots+(m-j)}}{(q; q)_j (q; q)_{m-j}} \quad (85)$$

$$= \frac{(-1)^{m-j} q^{\binom{m-j+1}{2}}}{(q; q)_j (q; q)_{m-j}}, \quad (86)$$

which is the raw q -Pochhammer form used throughout.

C Reproducing the computations

The archive contains:

File	Purpose
dyadic_up_extraction.tex	complete self-contained article source
dyadic_up_extraction.pdf	rendered article
dyadic_up_extraction.py	exact rational verifier and command-line tool
verification_output.txt	default exact verification run
README.txt	build and execution instructions

To rebuild the PDF with a standard TeX Live installation, run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error dyadic_up_extraction.tex
```

To repeat the exact experiments, run

```
python3 dyadic_up_extraction.py
python3 dyadic_up_extraction.py --x 1/16 --check-through 10
```

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